Market Risk Quantification: Value at Risk (VaR) via Monte Carlo Simulation on the BSE Sensex

Monte Carlo Methods in Finance

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1 Business Motivation and Regulatory Context

Risk management is exceptionally crucial in the banking sector due to the systemic role banks play in the broader economy. The risk management workflow strictly follows three steps: Identify, Quantify, and Mitigate.

A key regulatory metric is Leverage, defined as Capital divided by Total Assets. Under Basel II norms, this leverage ratio must be maintained at a minimum of 4%. Consequently, if a bank's losses exceed 4% of its total assets—meaning it fails to recover at least 96% of its loans or asset values—the institution is at risk of insolvency. Quantifying market risk using Value at Risk (VaR) allows banks to determine the exact amount of capital that must be set aside to safely support their trading positions and adhere to these strict regulatory frameworks.

2 Dataset

	Date	Open	High	Low	Close
1289	17-Mar-21	50436.02	50561.12	49718.65	49801.62
1290	18-Mar-21	50161.25	50296.35	48962.36	49216.52
1291	19-Mar-21	48881.19	50003.58	48586.93	49858.24
1292	22-Mar-21	49878.77	49878.77	49281.02	49771.29
1293	23-Mar-21	49876.21	50264.65	49661.92	50051.44

Figure 1: Dataset

The table presents historical daily OHLC market data for the index. Each row represents one trading day and includes:

- **Open** – Opening price
- **High** – Highest price during the day
- **Low** – Lowest price during the day
- **Close** – Closing price

For risk analysis, daily returns are calculated using consecutive closing prices, since the closing price reflects the final market value for the day. These returns are used to estimate historical volatility, which forms the basis for the Value-at-Risk (VaR) and Monte Carlo simulations.

3 Exploratory Data Analysis (EDA) and Fat Tails

Prior to simulation, an empirical analysis of historical index data (BSE Sensex, Jan 2016 - Mar 2021) was conducted. For VaR calculations, a common assumption is that asset

returns are normally distributed. However, empirical market data frequently violates this assumption.

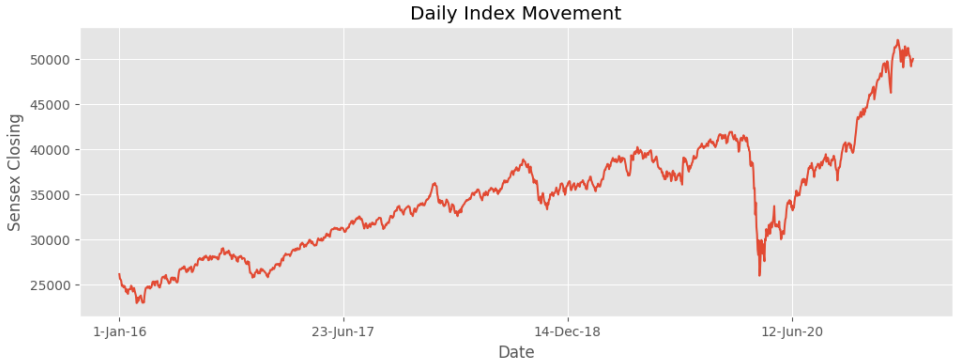


Figure 2: Daily Index Movement of BSE Sensex (2016–2021)

The first look at the index return shows that the daily Sensex returns are approximately normally distributed; however, there are outliers. Distribution of daily index returns are rather symmetric about the mean, but the tails are fatter—meaning more outliers—than what would be expected with the normal distribution.

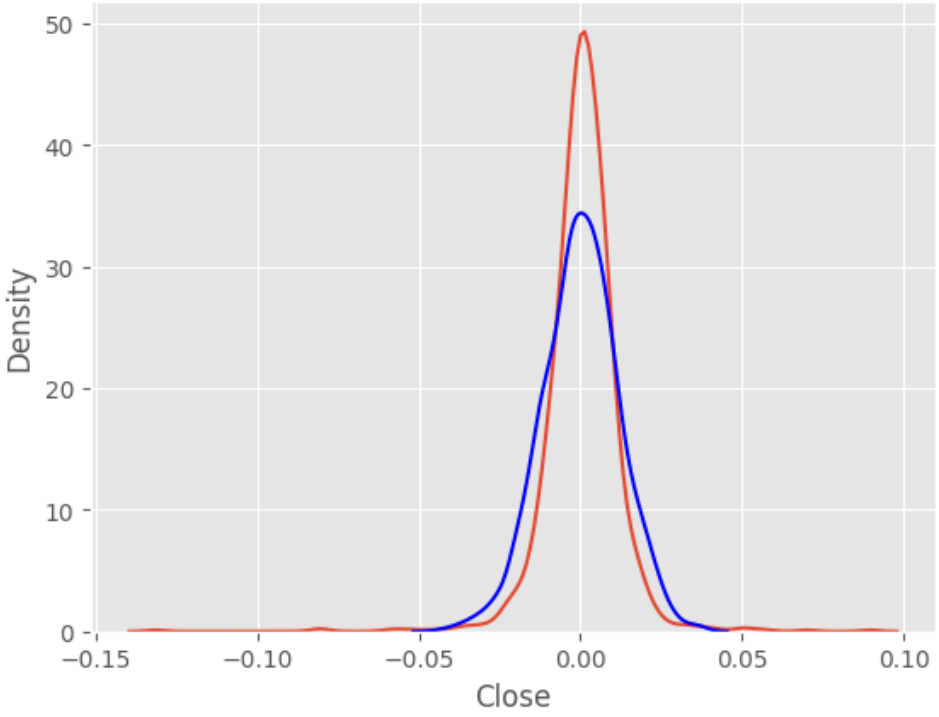


Figure 3: KDE Plot: Empirical Sensex Returns (blue) vs. Normal Distribution (red)

As seen in Figure 3, the peakedness (kurtosis) and the extreme values in the tails confirm that a standard normal distribution may under-represent the true risk during market stress events.

While the distribution of daily Sensex returns is roughly symmetric about its mean, Kernel Density Estimation (KDE) and Box Plots reveal that the empirical data exhibits

"fat tails." This indicates a significantly higher frequency of extreme outliers (e.g., daily returns exceeding ±5%, such as the March 2020 COVID-19 crash) than what a theoretical normal distribution predicts. Recognizing these fat tails is critical, as assuming pure normality in financial markets often leads to an underestimation of tail risk.

4 Theory: Monte Carlo & The Discrete Random Walk

Monte Carlo simulation is a computational algorithm that relies on repeated random sampling to obtain numerical results. To simulate future asset prices, we rely on stochastic calculus. The standard continuous-time model for asset prices is Geometric Brownian Motion (GBM):

$$dS_t = \mu S_t dt + \sigma S_t dW_t \tag{1}$$

where μ is the drift, σ is the volatility, and W_t is a Wiener process.

In our practical implementation, we assume a short-term horizon where the daily drift μ is negligible ($\mu \approx 0$). Using the discretization method, the continuous SDE is approximated into discrete daily steps as follows:

$$S_{t+1} = S_t(1 + \sigma Z) \tag{2}$$

where $Z \sim N(0, 1)$ is a standard normal random variable. This discrete random walk forms the mathematical basis of our simulation loop.

5 Algorithm & Methodology

Our approach combines Monte Carlo path generation with Parametric VaR evaluation. The Parametric VaR for a given position is defined as:

$$VaR = \text{Position} \times \sigma_{\text{daily}} \times Z_{1-\alpha} \tag{3}$$

The algorithm proceeds as follows:

1. **Data Extraction:** Calculate historical daily returns and extract the empirical daily volatility (σ).
2. **Path Simulation:** Generate $N = 1000$ independent random price paths over a 252-day horizon using the discrete random walk formula.
3. **Path-wise VaR Calculation:** For each simulated path, calculate the standard deviation of its simulated returns, and compute a discrete VaR for that specific path at a 95% confidence interval ($\alpha = 1.645$).
4. **Aggregation:** Compute the expected (mean) daily VaR and its standard deviation across all 1000 simulations.

6 Python Implementation

Below is the core implementation used to simulate the Sensex paths and compute the VaR.

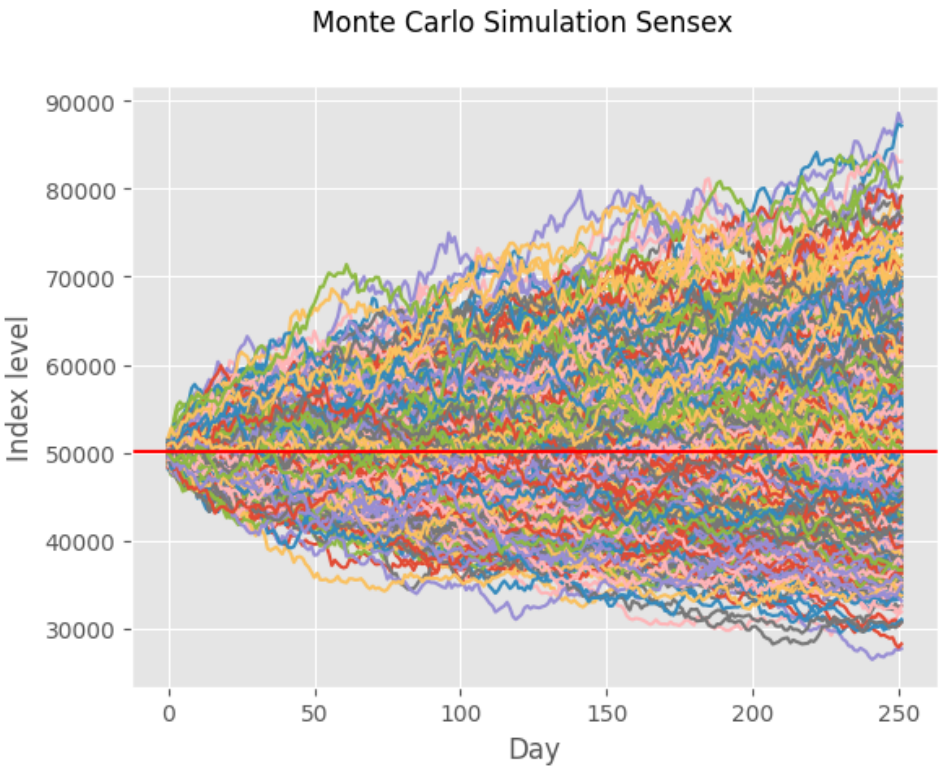


Figure 4: Monte Carlo Simulation of Sensex

7 Stress Testing

While historical daily volatility provides a baseline, robust risk management requires preparing for regime shifts. Stress testing answers the question: *How much additional capital must the bank raise if market volatility rapidly increases?*

Using the historical baseline, the annualized volatility was roughly 14.87%, yielding a baseline daily VaR of $\$1,911$. By artificially stressing the model, we observe the following capital requirements:

- **20% Annual Volatility Stress:** Translates to a daily volatility of $\sim 1.53\%$. The required VaR capital jumps to **$2,518**.
- **25% Annual Volatility Stress:** Translates to a daily volatility of $\sim 1.87\%$. The required VaR capital jumps to **$3,087**.

This non-linear increase in capital requirements highlights the sensitivity of trading portfolios to market shocks.

8 Conclusion

The techniques of stochastic simulation are invaluable in generating and testing hypotheses about market data. By modeling the BSE Sensex using a discrete random walk and conducting subsequent stress tests, we quantified the exact capital a bank must reserve to fund a $100,000 position under normal and stressed market regimes. This methodology directly bridges the gap between theoretical asset pricing and practical, regulatory capital allocation.